

Enriched Population Analysis

METIMMOX-1 Economic Evaluation

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1 Overview

This report presents an enriched population cost-effectiveness analysis (ceiling scenario) for the METIMMOX-1 biomarker-guided immunotherapy strategies. In the base case analysis, each biomarker strategy's costs and effects are calculated as a prevalence-weighted average of the biomarker-positive subgroup (receiving experimental treatment) and the biomarker-negative subgroup (receiving standard treatment). This dilutes the incremental benefit with patients who do not receive immunotherapy.

The enriched population analysis models a **biomarker-enriched trial design** where only biomarker-positive patients are enrolled. For each biomarker:

- **Enriched experimental arm:** biomarker-positive patients receive FLOX + nivolumab
- **Enriched control arm:** biomarker-positive patients receive FLOX only (counterfactual)

The enriched control curves are generated by predicting from the fitted survival models for biomarker-positive patients assigned to control treatment. Both arms incur the biomarker test cost. This provides an apples-to-apples comparison within the enriched cohort.

1.1 Biomarker Prevalences

Table 1: Biomarker-Positive Prevalences in the METIMMOX Trial

Biomarker	Prevalence	N Positive
CRP (C-reactive protein)	33.8%	24
TLR (Tumor lesion reduction)	64.7%	44
TMB/BRAF (TMB or BRAF mutation)	44.9%	31

1.2 Model Structures

Each enriched analysis is run under three alternative survival model specifications:

Table 2: Survival Model Specification

	Model	Description
Model_A	Model A: Joint	All biomarkers + all treatment int
Model_B	Model B: Focused	CRP/TMB_BRAF with both inte
Model_C	Model C: Separate	One biomarker + its interaction on

2 Base Case vs Enriched Population ICERs

The following tables compare the base case (population-weighted, vs overall control) and enriched (biomarker-positive only, vs biomarker-positive control) cost-effectiveness results. Base case ICERs compare each biomarker strategy to the overall control. Enriched ICERs compare biomarker-positive patients on experimental treatment to biomarker-positive patients on FLOX only.

Table 3: Pairwise ICERs vs Control: Base Case vs Enriched Population

Biomarker	Prevalence	Model	Base Case ICER	Enriched ICER	Change
CRP	33.8%	A	Dominated	Dominated	–
		B	Dominated	Dominated	–
		C	Dominated	Dominated	–
TLR	64.7%	A	Dominated	Dominated	–
		B	Dominated	Dominated	–
		C	Dominated	EUR 1,242,396	–
TMB/BRAF	44.9%	A	EUR 1,949,087	EUR 1,881,987	-3.4%
		B	EUR 14,980,066	Dominated	–
		C	EUR 696,842	EUR 323,878	-53.5%

Interpretation: A negative percentage change indicates the enriched population is more cost-effective than the base case. Strategies that are “Dominated” in the base case (higher cost and lower effect than control) may become non-dominated in the enriched population if the biomarker-positive subgroup has better treatment outcomes.

3 Detailed Cost and QALY Breakdown

3.1 Model A: Joint

Overall control (base case): Cost = EUR 20,891, QALYs = 1.846

Table 4: Enriched Cohort Results: Model A: Joint (Biomarker+ Experimental vs Biomarker+ Control)

Biomarker	Biomarker+ Experimental		Biomarker+ Control		Incremental	
	Exp. Cost	Exp. QALYs	Ctrl Cost	Ctrl QALYs	Inc. Cost	Inc. QALYs
CRP	EUR 121,691	2.369	EUR 21,144	2.553	EUR 100,547	-0.185
TLR	EUR 120,701	2.088	EUR 22,356	2.344	EUR 98,344	-0.256
TMB/BRAF	EUR 112,889	1.992	EUR 23,521	1.945	EUR 89,369	0.047

3.2 Model B: Focused

Overall control (base case): Cost = EUR 21,261, QALYs = 1.940

Table 5: Enriched Cohort Results: Model B: Focused (Biomarker+ Experimental vs Biomarker+ Control)

Biomarker	Biomarker+ Experimental		Biomarker+ Control		Incremental	
	Exp. Cost	Exp. QALYs	Ctrl Cost	Ctrl QALYs	Inc. Cost	Inc. QALYs
CRP	EUR 121,746	2.337	EUR 22,347	2.873	EUR 99,399	-0.536
TLR	EUR 122,839	2.082	EUR 22,870	2.353	EUR 99,969	-0.271
TMB/BRAF	EUR 107,775	1.886	EUR 24,265	2.191	EUR 83,510	-0.306

3.3 Model C: Separate

Overall control (base case): Cost = EUR 21,261, QALYs = 1.940

Table 6: Enriched Cohort Results: Model C: Separate (Biomarker+ Experimental vs Biomarker+ Control)

Biomarker	Biomarker+ Experimental		Biomarker+ Control		Incremental	
	Exp. Cost	Exp. QALYs	Ctrl Cost	Ctrl QALYs	Inc. Cost	Inc. QALYs
CRP	EUR 122,925	2.366	EUR 22,180	2.858	EUR 100,745	-0.492
TLR	EUR 125,779	2.247	EUR 22,415	2.164	EUR 103,364	0.083
TMB/BRAF	EUR 122,529	2.146	EUR 24,148	1.842	EUR 98,381	0.304

4 Cost-Effectiveness Planes

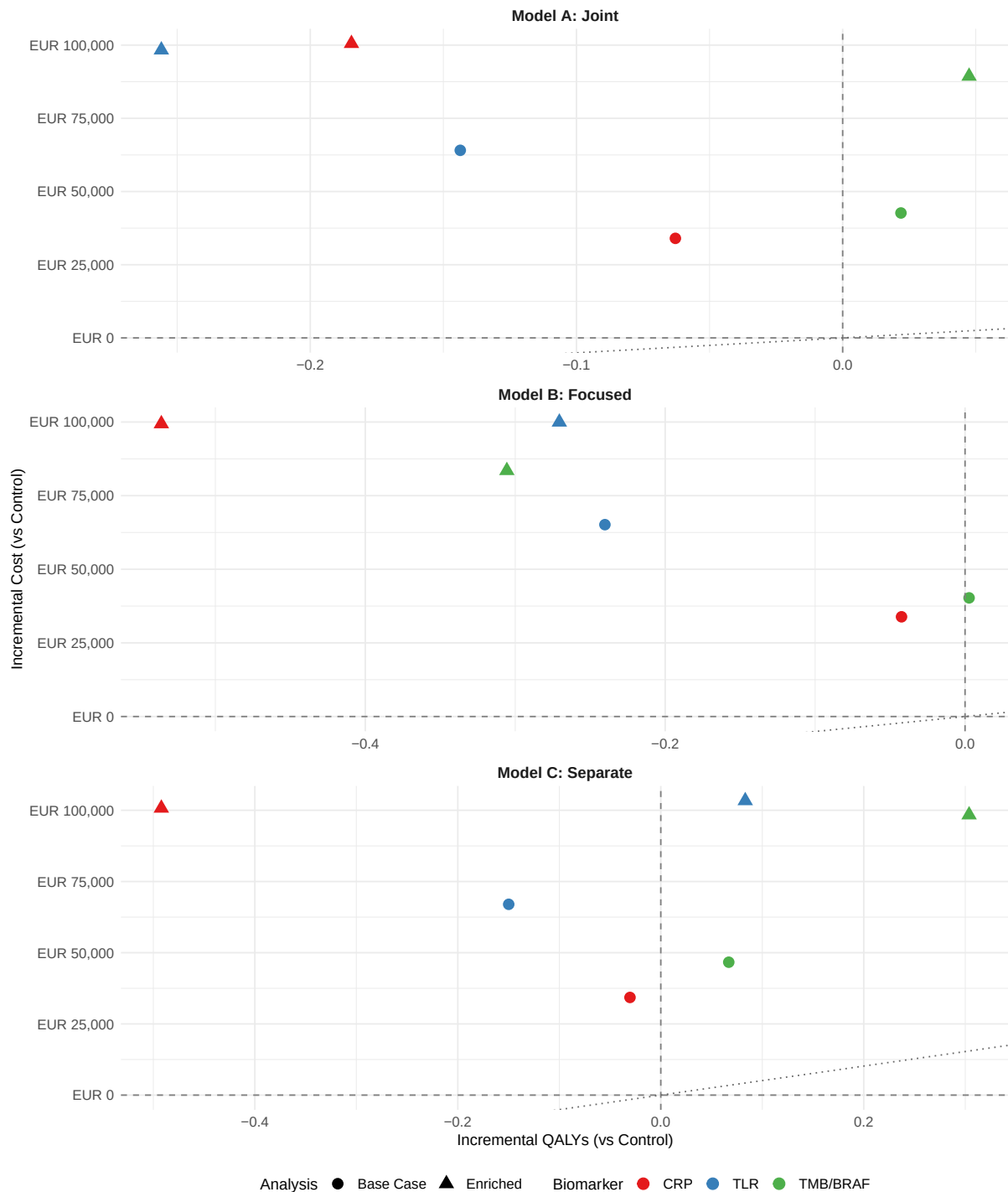


Figure 1: Incremental Cost-Effectiveness: Base Case vs Enriched Population

The dotted line represents the willingness-to-pay threshold (EUR 51,000/QALY). Points below this line are cost-effective. Base case points (circles) show incremental cost/effect vs the overall control population. Enriched points (triangles) show incremental cost/effect vs the biomarker-positive control population. The enriched comparison uses a matched control arm (same biomarker-positive patients on FLOX only), providing an apples-to-apples comparison.

5 Interpretation

5.1 Mechanism of ICER Improvement

In the base case, biomarker strategy costs and effects are prevalence-weighted averages:

$$\text{Strategy Cost} = p \times \text{Cost}_{pos} + (1 - p) \times \text{Cost}_{neg}$$

where p is the biomarker prevalence. The biomarker-negative subgroup receives standard treatment (same as control), so their incremental cost and effect are near zero. This dilutes the incremental benefit from the biomarker-positive subgroup.

In the enriched analysis, only biomarker-positive patients are considered in **both** the experimental and control arms. The enriched control arm uses survival curves predicted from the fitted models for biomarker-positive patients assigned to FLOX only. This removes dilution from biomarker-negative patients and provides a direct comparison of treatment effect within the target population.

5.2 Clinical Implications

Table 7: Summary of Enrichment Effects on Cost-Effectiveness

Biomarker	Prevalence	Enrichment Effect (Model A)
CRP	33.8%	Base: Dominated; Enriched: Dominated
TLR	64.7%	Base: Dominated; Enriched: Dominated
TMB/BRAF	44.9%	ICER improves by 3% from EUR 1,949,087 to EUR 1,881,987

5.3 Limitations

- This is a deterministic analysis only; probabilistic sensitivity analysis of the enriched scenario is not included.
- The enriched control arm survival curves are counterfactual predictions from the fitted model (biomarker-positive patients assigned to control treatment), not observed data from a separate enriched trial.
- The analysis assumes perfect biomarker test sensitivity and specificity.

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